Green for the Ambulance at Perempatan Fatmawati: A Digital Twin for Learning Emergency-Vehicle Signal Priority in Motorcycle-Dominated Traffic

ITI5145 Foundations of Data Science, Term 4 2026 — Assignment 1: Project Proposal

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Business scenario: Healthcare (emergency medical services)

1. Introduction

1.1 Problem and context

Perempatan Fatmawati, under South Jakarta's JORR toll viaduct, lies 326 m north of RSUP Fatmawati, a national referral hospital with a 24-hour emergency department (Direktorat Jenderal Kesehatan Lanjutan, n.d.; Figure 1). About four in five vehicles here are motorcycles (Yusuf & Isradi, 2025); at red they fill the motorcycle box and every gap between cars (Figure 2). Indonesian law gives ambulances right of way and exempts them from signals (Undang-Undang Nomor 22 Tahun 2009, arts. 134–135), yet a boxed-in queue cannot move aside: only a timely green for the ambulance's approach clears the path. The junction already runs at the worst level of service, averaging 85 s delay per vehicle (Sumabrata et al., 2026).

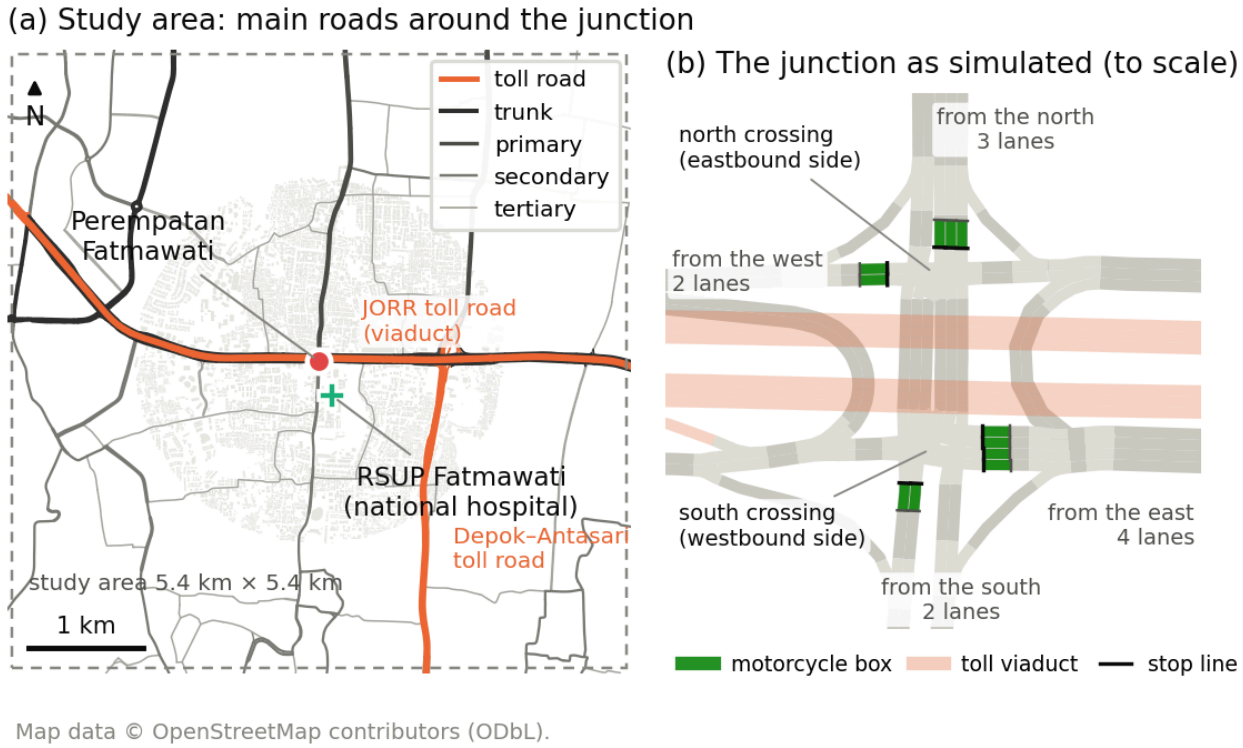


Figure 1. The study area in a 5.4 km square (a) and the junction as simulated (b).

Problem statement. No data-driven ambulance-priority rule exists for motorcycle-dominated junctions, because ambulance passages cannot be recorded safely on live roads. We will build a digital twin of the

junction, generate open sensor data on ambulance passages, and learn a priority controller that cuts ambulance delay while keeping every direction's delay fair and bounded.

1.2 Why it matters

Survival from cardiac arrest falls by about two percentage points for every minute advanced life support is delayed (Larsen et al., 1993). Among Jakarta emergency-department patients, the median ambulance response was 24 minutes and only 9% arrived by ambulance (Brice et al., 2022).

1.3 Goals

1. **Twin.** Build the junction and surrounding main roads in SUMO from OpenStreetMap, with motorcycle boxes, the city's four-stage signal plan, and drivers yielding to ambulances only where there is room.
2. **Dataset.** Publish at least 100 reproducible simulated hours as CSV tables of virtual sensor readings (loops, queue cameras, counting gates, ambulance tracker), with a data dictionary and card.
3. **Controller.** Train model families from logistic regression to gradient boosting and a small neural network to imitate a priority rule that requests the ambulance's stage once its arrival time falls within the time needed to clear the queue ahead:
$$\mathrm{ETA} \le I + l_s + \frac{3600}{S} \sum_k e_k \, n_k + \Delta,$$
where I is the intergreen, l_s the start-up lost time, S the saturation flow, n_k the vehicles of class k ahead and e_k their car equivalents (Direktorat Jenderal Bina Marga, 2023), plus Δ, the 5 s decision interval. Labels are verified by recomputation; models must beat stillness and random baselines and only request stages, which the signal plant applies safely.
4. **Evaluation.** Test every method on one grid of ambulance cases against an empty-road reference: ambulance delay averaged with equal weight per direction, worst direction shown, each approach's delay indexed to the city's plan; Nash bargaining (Nash, 1950) picks the most balanced method.
5. **Live view.** Show live TomTom speeds beside the twin; the licence permits display only.

2. Related Work

2.1 Traffic signal control

Fixed-time plans follow Webster (1958), adapted for Indonesia's mixed traffic (Direktorat Jenderal Bina Marga, 2023). Max-pressure control (Varaiya, 2013) uses only local queues. Deep reinforcement learning dominates research but is hard to audit (Wei et al., 2019); on tabular data, tree ensembles match deep models and stay interpretable (Grinsztajn et al., 2022).

2.2 Emergency-vehicle priority

Chen et al. (2024) time preemption around queue spillback; Roy et al. (2024) tune it in a Vissim-based digital twin; EMVLight (Su et al., 2023) treats yielding as a speed rule. Savithramma et al. (2022) predict queue-clearing green time with gradient-boosted trees, but they, like a Surabaya SUMO study (Izzuddin et al., 2025), treat two-wheelers only as counts.

2.3 Motorcycles, simulation and twins

Motorcycle boxes shorten motorcycles' discharge headways (Wichitphongsa et al., 2025); SUMO (Alvarez Lopez et al., 2018) represents filtering with its sublane model (Semrau & Erdmann, 2016). A

transport digital twin joins a model, a live data link and synchronisation (Kušić et al., 2023); like other SUMO twins (Schaffland et al., 2024), ours defers synchronisation: TomTom's terms forbid storage and safety-critical use (TomTom, n.d.).

2.4 Gap and novelty

- **A real Jakarta junction beside a national hospital,** with motorcycle boxes, left-hand driving and the city's own signal plan.
- **Space-limited yielding:** drivers move aside only into real gaps.
- **An open dataset** of ambulance passages as sensor tables with verified labels, which we have not found elsewhere.
- **Interpretable control behind a safety shield,** scored over every ambulance case, neglecting no direction.

3. Business Model

3.1 Application area and the role of data

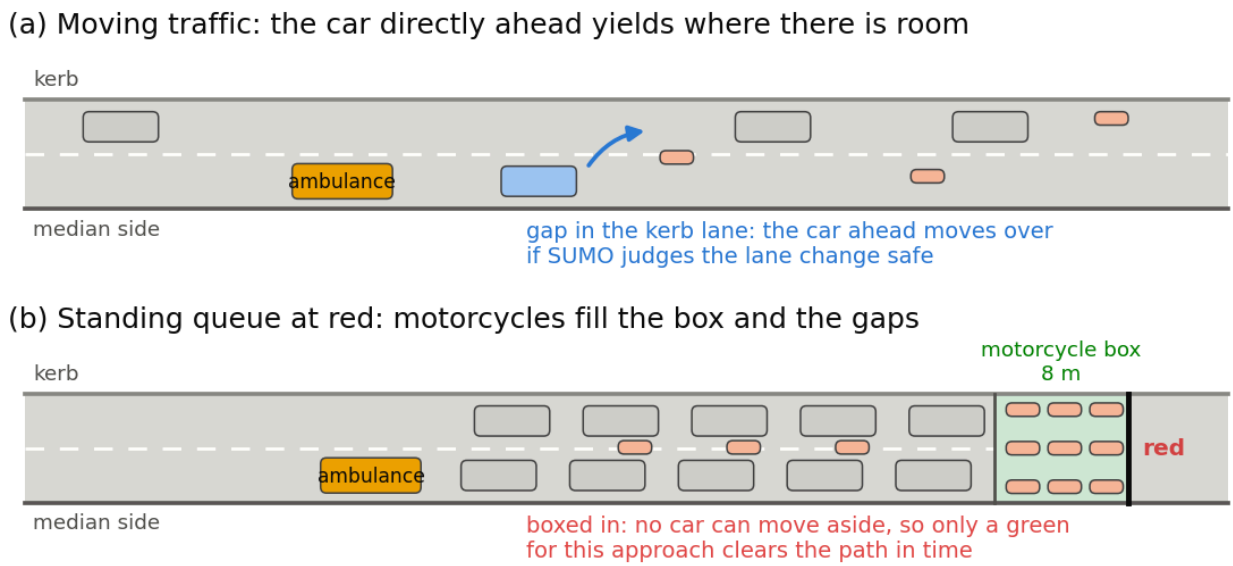


Figure 2. Drivers yield only where there is room: moving traffic can (a); a queue at red cannot (b).

Three public bodies share the problem: RSUP Fatmawati's emergency department, Jakarta's 119 ambulance service (Pusat Krisis dan Kegawatdaruratan Kesehatan Daerah, n.d.) and the Transport Agency's signal-control unit, UP.SPLL. Data flow in three tiers: OpenStreetMap geometry (ODbL); published counts, signal plan and queues for calibration (Sumabrata et al., 2026); and the twin's sensor tables, the product: training data whose inputs real detectors deliver, investment evidence and a reusable benchmark. Preprocessing: warm-up removal, label recomputation, episode-grouped splits; demonstration in R, models in Python.

3.2 Value and benefits

For patients and crews, seconds saved and stops avoided. For the Transport Agency, a bounded cost: general delay within a proposed 10% of max-pressure, no approach more than 20% worse than today,

recovery reported. For the city, evidence before spending and a pipeline that rebuilds any junction from open maps.

3.3 Stakeholders and their needs

Stakeholder	Need	How the project meets it
Patients and ambulance crews	Fewer stops, predictable passage	Primary measure; pre-clearing
RSUP Fatmawati emergency department	Earlier, predictable arrivals	Passage times for every direction
UP.SPLL signal operators	No manual intervention, no gridlock	Shielded requests; gridlock counted
Motorcyclists and other road users	Fair, bounded delay	Approach delay against today; worst shown
City government and researchers	Evidence, reusable data	Paired-seed tests; open tables, data card

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Use of generative AI

We used Claude (Anthropic, via Claude Code, September–October 2026) to brainstorm, critique the simulator design, draft and revise this proposal from our decisions and notes, draft code, label checks and figures that we reviewed and tested, and check references by web search. The problem, decisions and sources are ours; we re-checked every reference.